

Exit Ticket: Inverse Trigonometric Functions — Key

1. Evaluate each expression. Give the exact answer in radians.

(a) $\arccos\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$

(b) $\arctan(-\sqrt{3}) = -\frac{\pi}{3}$

2. Explain why the domain of sine must be restricted.

The sine function is periodic and many-to-one: the same output value is produced by infinitely many input angles. Because it fails the horizontal line test, it does not have an inverse unless we restrict its domain to an interval on which it is one-to-one, such as $[-\frac{\pi}{2}, \frac{\pi}{2}]$.